



SCIENTIFIC SUPPORT

Criteria used to set each critical limit

Ready-to-eat, heat-treated, shelf-stable beef jerky

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This is the support requested for the critical control points. For each critical limit it names the hazard, the number, the criterion used to choose that number, and the page where that criterion is written. It is written so the reviewer can see why each limit is the limit for this ready-to-eat product.

The product is whole-muscle beef jerky cooked in this plant's gas oven and then dehydrated in that same oven. The oven vent stays open. The jerky is not fermented, is not acidified, and contains no nitrite. Nothing is added after the heat step. pH is not a critical limit. Soy and wheat are in the formula and are declared on the bag. They are controlled by the batch sheet and the label check. They are not a critical control point.

A lot is released on the process records. This support does not rely on a finished-product laboratory test, and it does not cite a challenge study of this jerky. 9 CFR 417.4(a)(1) still has an in-plant part. The validation date on the HACCP plan stays blank until logs from this oven, this load, and this thickness have been reviewed.

1. The process the limits belong to

Stage	What the plant does	Why it is in this order
CCP 1, lethality	Strips are sealed in a moisture-impermeable bag and heated in the gas oven until the internal temperature is 160°F or above, with a come-up of 6 hours or less between 50°F and 130°F.	Appendix A says humidity has to be applied during lethality, before drying (page 26). The sealed bag is the way this oven keeps the surface moist (page 31).
CCP 2, dehydration	The bag is opened only after CCP 1 passes. The gas oven then dries the strips at a dry bulb of 170°F or above until water activity is 0.85 or less.	Drying is what makes the jerky shelf-stable. It is not the lethality step.

2. CCP 1 — lethality

2.1 Internal temperature 160°F or above

- Hazard. Salmonella and Shiga-toxin producing Escherichia coli, including O157:H7, on raw beef, including on the surface of a sliced strip. Listeria monocytogenes is included in the cook because the product is ready to eat.
- Critical limit. Internal temperature 160°F or above, in the thickest piece, in the coldest part of the load. Readings are not averaged. The oven-air temperature is not the limit.
- Criterion. FSIS Cooking Guideline for Meat and Poultry Products (Revised Appendix A), December 2021, FSIS-GD-2021-14, Table 2, page 35, “Time-Temperature Combinations for Meat Products to Achieve Lethality.” Footnote 5 on that page: the required log reductions are achieved instantly (0 seconds) when the internal temperature of a cooked meat product reaches 158°F or above. The row at 145°F on the same table is 4 minutes for both the 6.5-log column and the 7-log column. This plan does not use the 145°F row. It uses 160°F, which is above the 158°F instantaneous line, so the dwell time at 160°F is zero on that table.
- The same table states that relative humidity and come-up time are also critical operating parameters when the table is used. Those two parameters are the next two limits. They are not optional.
- The preservation chapter, page 167, says processors must validate a 5-log reduction of E. coli O157:H7 for products that contain beef, and that the product must be heated to achieve it. The heat step in this plan is the Table 2 endpoint above.

2.2 Come-up time, 6 hours or less

- Hazard. Staphylococcus aureus growth and heat-stable enterotoxin if the meat stays too long between 50°F and 130°F during heating.
- Critical limit. The time the internal temperature is between 50°F and 130°F is 6 hours or less.
- Criterion. Appendix A, page 24: “Come-Up-Time Option: Total time product temperature is between 50 and 130°F is 6 hours or less.” The same page says those temperatures are internal temperatures. Footnote 7 of Table 2, page 35, repeats the 6-hour limit.
- The meat is held at or below 41°F before it is cooked, so it enters the oven below 50°F. The clock starts when the probed piece reaches 50°F and stops when it reaches 130°F.

2.3 Sealed moisture-impermeable bag

- Hazard. Salmonella surviving on a surface that dries before the lethal internal temperature is reached. Drying the surface early makes the organism harder to kill.
- Critical limit. The strip stays in a sealed moisture-impermeable bag until the internal temperature is 160°F or above. The bag is opened only after that reading is recorded. Dehydration starts after that.
- Criterion. Appendix A, page 31, “Situations when Humidity is Not Needed.” Moisture is inherently maintained when the product is cooked in a sealed, moisture-impermeable bag. Establishments that match that situation do not monitor relative humidity as its own critical operating parameter. The bag is that situation for this gas oven.

- Appendix A, page 26: humidity is applied during the lethality treatment, before drying. Opening the bag only after 160°F puts dehydration after lethality.
- Appendix A, page 29: if an oven has an opening that cannot be closed, the sealed-oven method is not used. This is a gas oven. The burner vent stays open. The plan therefore does not use Humidity Option 2.
- Appendix A, page 27: a wet-bulb temperature of 125–130°F and 27–32 percent relative humidity for one hour, taken from the jerky guideline, is not adequate on its own to support the Appendix A humidity options. This plan does not use that pair as the humidity limit. The jerky checklist prints a wet bulb of 125–142°F next to its heat-lethality line. That line is not a critical limit in this plan, for the reason on page 27.
- The preservation chapter, Jerky Products section, page 168, says the jerky guidelines call for a humidity step at the beginning of the process. The sealed-bag cook is that step. Dehydration follows it.
- The same chapter, page 166, states that drying or a lower water activity can increase heat resistance, and that the heating medium, including water activity, changes that resistance. Cooking while the strip is still moist is the control for that increase. The surface is not dried first.

3. CCP 2 — dehydration

3.1 Dry bulb 170°F or above

- Hazard. Outgrowth of *Clostridium perfringens* and *Clostridium botulinum* while water activity is still high enough for growth, during the drying stage.
- Critical limit. The dry-bulb temperature of the gas oven is 170°F or above throughout drying, read before the jerky is removed.
- Criterion. The jerky field checklist, heat-lethality row, prints a dry bulb above 77°C (170°F). This plan applies that dry-bulb temperature to the whole drying stage. The FSIS HACCP Model for Ready-to-Eat, Heat-Treated, Shelf-Stable Beef Jerky, 2021-0004, CCP 2, sets the oven at 170°F or above during drying so those sporeformers do not grow while water activity is still above 0.93, and it cites the FSIS Stabilization Guideline, Revised Appendix B, December 2021, for that temperature. Lethality itself is CCP 1, not this dry-bulb reading.

3.2 Water activity 0.85 or less

- Hazard. *Staphylococcus aureus* growth and toxin on a jerky that is stored in air. *Listeria monocytogenes* growth during storage. Growth of the vegetative pathogens named in the preservation table if water activity stays above their minima.
- Critical limit. Finished water activity 0.85 or less. At least six pieces from different places in the lot, including a thick piece. The highest reading is the lot result.
- Criterion. Principles of Preservation of Shelf-Stable Dried Meat Products, October 31, 2011, growth-minimum table (chapter pages 157–158): *Staphylococcus aureus*, aerobic, 0.85. The same

chapter, page 160: *Staphylococcus aureus* can grow as low as 0.86 water activity when oxygen is present. A finished result of 0.85 or less is at or below that aerobic minimum and below 0.86. The same table: *Escherichia coli* O157:H7, 0.95; *Salmonella*, 0.94; *Listeria monocytogenes*, 0.92; *Clostridium perfringens*, 0.93; proteolytic *Clostridium botulinum*, 0.93. Finished jerky at 0.85 or less is below each of those minima.

- The same chapter, page 160: dried hams, coppa, and beef jerky generally have water activity less than 0.88. The jerky field checklist, drying row, prints water activity below 0.88. The limit in this plan is 0.85 or less, which meets the checklist.
- The same chapter, Jerky Products section, page 168: safety and shelf stability of jerky are judged by water activity, not by the moisture-to-protein ratio, and the older practice of relying on a moisture-to-protein ratio of 0.75 or below is not the safety indicator. Page 160 states that moisture-to-protein ratios are labeling standards and are not necessarily indicative of microbial safety. This plan checks 0.75:1 on validation lots for the product name. It is not a critical limit.
- The chapter also states that if pathogens are still viable, the product is adulterated. Water activity stops growth. It is not the kill step. The kill step is CCP 1.
- The growth table lists *Staphylococcus aureus* at 0.90 when oxygen is absent and at 0.85 when oxygen is present. This plan does not claim an oxygen-free package, so the aerobic number is the one used. The same table lists non-proteolytic *Clostridium botulinum* at 0.96. Finished jerky at 0.85 or less is below that minimum as well.

3.3 How this chapter applies to this jerky

Principles of Preservation of Shelf-Stable Dried Meat Products, October 31, 2011, is support for the drying limit and for the decision not to copy other dried-meat processes into this plan. The parts that apply, and the parts that do not, are named here.

- Two preservation types from page 157 apply. Heat inactivates pathogens: CCP 1. A low water activity then slows or stops growth: CCP 2. Page 157 says water activity is the most important single factor for shelf stability across dried meats. Page 156 says shelf life is set for acceptable quality, because safety is addressed in the process. The blank shelf-life line in the HACCP plan is that quality decision. It is not a critical limit. The 70°F line on the bag is a storage instruction, not the safety limit.
- Page 166 requires a validated 5-log reduction of *E. coli* O157:H7 in products that contain beef, and says that reduction is, for the most part, achieved by heating. Page 166 also says a published validation may be used when it is equal to or less severe than the process being validated. The published lethality support for this jerky is Appendix A, Table 2, not the summer-sausage, pepperoni, or country-ham examples later in the chapter. Those examples are different products.
- Page 166 says pathogenic sporeformers may need to be addressed. While the strips are still moist, the dry bulb of 170°F or above is that address. The same page says that once water activity is low, pathogenic sporeformers should not be a concern in the finished product. Finished water activity of 0.85 or less is below the table minima of 0.93 for *Clostridium perfringens* and proteolytic *Clostridium botulinum*.

- Page 158 says that, for many shelf-stable dried meats, acidity, low redox potential, preservatives, and competitive flora are the main hurdles, because those products are often not pasteurized. This jerky is heat-treated. Those four hurdles are not controls in this formula. Vinegar in the marinade is not measured as a pH limit and does not replace the cook or the water-activity limit.
- Not used, because this is not that process. Degree-hours to pH 5.3, pages 163–164, apply to fermented sausage with a starter culture. This jerky has no starter culture and no fermentation step. The salting step for dry-cured whole muscle, pages 164–165, is not this process. Salt in the marinade is for flavor. Brine strength is not a critical limit. Nitrate, nitrite, and cure accelerators, pages 160–161, are not in the formula. The pH minima on page 158 are not a critical limit. This plan does not acidify the meat to a target pH.
- Mold is not stopped by a water activity of 0.85. The growth table gives *Aspergillus flavus* a minimum of 0.80. Pages 162–163 say mold can grow on dried meat that is not in an anaerobic package, and that mold can raise the pH. This plan does not add a mold inhibitor and does not claim vacuum or modified atmosphere. The HACCP plan controls that as a prerequisite: the bag is heat-sealed, stored cool and dry, and a bag with visible mold does not ship. It is not a third critical control point.

4. Limits that are not critical control points

Item	Number used	Why it is not a CCP
Cooler	Raw meat at or below 41°F (5°C).	Jerky field checklist, cooler-storage row. It keeps the meat below the start of the come-up range. Outgrowth before the cook is controlled by this prerequisite.
Employee health	Restriction and exclusion under the 2022 FDA Food Code, part 2-201, using the Employee Health Policy Tool dated August 23, 2023.	People who are ill do not handle the product. The tool does not identify a process critical limit.
Allergens	Contains: Soy, Wheat, on every bag. Formula in the plan, Section 3.1.	Controlled by the batch sheet and the pack-out check.
Moisture-to-protein ratio	0.75:1 or less, on validation lots.	Standard of identity for the name jerky. Preservation chapter, page 160 and page 168. Not a safety limit.
pH, fermentation, degree-hours	Not used.	Preservation chapter, pages 158 and 163–164. This jerky is not fermented and is not acidified to a target pH.
Salt and brine strength	Not a safety limit.	Preservation chapter, pages 160 and 164–165. Those controls are for cured and dry-cured products. Salt here is for flavor.
Mold	No visible mold. The bag stays sealed.	Preservation chapter, pages 162–163, and <i>Aspergillus flavus</i> at 0.80. Prerequisite. Not a critical control point.
Curing and smoking checklist	Not used.	That checklist is for brine, dry cure, a salinometer, and hanging meat before smoke. This jerky is not that process. Using it would attach the wrong criteria to these critical limits.

5. What the returned May plan did not show

The returned plan set CCP 1 at 160°F and CCP 2 at a water activity of 0.85 or less. It did not identify the Table 2 footnote that makes 158°F instantaneous, it did not set the 6-hour come-up, and it did not state how moisture is maintained in this gas oven. Section 2 of this document is that identification. The water-activity number is kept and is now tied to the preservation-chapter table and to the checklist value it also satisfies.

6. Signature

I am providing this document as the scientific support for the critical limits in the HACCP plan for this ready-to-eat beef jerky.

Responsible establishment official, Jesús Canales: _____

Date: _____